AFM 425 Fixed Income Securities

Fall 2026

Assignment #1

Submission Deadline: 11:59 PM Friday, October 2, 2026

Instructions

Content

- This assignment covers Module 1 to 3 of the course.
- It includes four Problem Solving Cases, each worth 25 points.

Responses

- For time-value-of-money calculations, include all inputs in your responses (e.g., N=10, PMT=2.5, etc.). You can use Excel, analytical formula, or Python to perform the actual calculation.
- You are permitted to use internet resources or consult other materials to complete this assignment. Please provide adequate references where necessary.
- Problem Solving Cases 1 to 3 consist of questions with definitive answers, designed to assess your understanding of the course materials. Problem Solving Case 4 contains open-ended questions that encourage research and personal opinion, aimed at evaluating your ability to gather and analyze information in real-world contexts.

Submission

- Deadline: 11:59 PM, Monday, October 2, 2026.
- Method: Submit via the LEARN Dropbox.
- Required files: One PDF document with your responses and one Excel file with your calculations and the amortization table.
- This is an individual assignment. Ensure that each submission reflects your own work.

Problem Solving Case #1 – Bond Valuation (25 points)

You are an investment advisor, and one of your clients, Mrs. Smith, is always eager to learn. Your meetings with her often become educational sessions. During a recent portfolio review, Mrs. Smith expressed a keen interest in understanding fixed-income investments.

1) You discuss a fixed-coupon corporate bond with a quoted annual YTM of 6%. Mrs. Smith believes that by purchasing the bond, she will receive a return of 6%. How would you clarify the concept of YTM to her, and under what conditions would she realize a yield-to-maturity of 6%? (6 marks)

2) You then examine a bond issued by Heavy Equipment Inc. (referred to as the "Heavy Equipment Bond") set for settlement on July 3, 2026. Information on the Heavy Equipment Bond is as follows:

Annual Coupon	9%
Coupon Frequency	Semi-Annual
Interest Payment Dates	June 30 and December 31
Maturity Date	December 31, 2032
Day-Count Convention	30/360
Annual Yield-to-Maturity	7.6%
Par Value	$100

What would be the full price of the Heavy Equipment Bond on the settlement date? (6 marks)

3) In line with bond market practices, how would a dealer quote the price of this bond on the settlement date? Explain to Mrs. Smith the difference between the quoted price and the full price on the settlement date. Calculate accrued interest and the clean price (7 marks)

4) 4) Mrs. Smith spots another bond from Rock Corp. (the "Rock Bond") with similar terms but a stated annual yield based on monthly compounding of 8.6%. She thinks the Rock Bond is less attractive than the Heavy Equipment Bond due to its lower yield. How would you guide her in comparing the two bonds? (6 marks)

By the end of the session, Mrs. Smith is pleased with her newfound knowledge and your investment expertise. Your client relationship is strengthened, and your investment portfolio continues to grow.

Problem Solving Case #2 – Bond Amortization (25 points)

Suppose you are given the following U.S. corporate bond information:

Issuer:	Advanced Automated Manufacturing
Coupon Rate:	6.50%
Maturity date:	2036-08-15
Day count convention:	30/360
Par Value	$100

The yield-to-maturity of the bond is 9%, and the settlement date for purchasing the bond is August 15, 2026.

1) Calculate the bond price as of the settlement date. (6 marks)

2) 2) Determine whether the bond is considered a premium bond or a discount bond. (4 marks)

3) Create an amortization table for this bond. You should construct the table in Excel using appropriate formulas. Additionally, provide a written explanation of the process involved in creating this table. (15 marks)

Problem Solving Case #3 – Yield Curve Bootstrapping (25 points)

Assume the following government treasury securities are observed from the market.

Bond	Coupon Rate	Maturity	Price
1	3.75%	6-month	99.177975
2	5.05%	1-year	99.399571
3	6.39%	1.5-year	100.547780
4	2.44%	2-year	94.424926
5	7.30%	2.5-year	105.038305
6	6.47%	3-year	104.589915

1) Using the bootstrap method discussed in this course, construct the zero-coupon yield curve. Ensure that your yield curve includes the same tenors as those listed in the table above. (8 marks)

2) 2) In accordance with the pure expectations hypothesis, what will be the price and yield-to-maturity (YTM) of a 5% coupon Treasury note with a 2.5-year term to maturity, six months from now? (5 marks)

3) 3) Calculate the price of a corporate bond with a 6% coupon rate and 3 years to maturity, currently priced off the yield curve with a z-spread of 150 bps. (5 marks)

4) 4) Consider a hypothetical bond issued by Scientific Health Solutions Inc. with the following characteristics. Determine the z-spread and nominal spread of the bond. (7 marks)

Issuer:	Scientific Health Solutions Inc.
Coupon rate:	5.35%
Term to maturity:	3 years
Day count convention:	30/360
Market Price:	95.45
Benchmark yield:	4.50%

Problem Solving Case #4 – Term Structure of Interest Rates (25 points)

In this course, we have discussed the various shapes of yield curves. This question extends that discussion.

1) Using data provided by the U.S. Department of the Treasury (link is provided below), plot yield curves for selected dates sampled from the following periods (one date for each period). Ensure that your chosen dates effectively represent the respective periods. Analyze and explain your observations from the plots. (15 marks)
    a. The Global Financial Crisis (GFC).
    b. Pre-COVID period.
    c. Around April 20, 2020 (This period marks the height of the COVID panic, characterized by the sub-zero crude oil prices).
    d. Prior to the 2022/23 rapid rate hikes.
    e. Period between the recent rate hikes and now when the curve is still inverted
    f. The current period.

2) The inverted yield curve is often considered an indicator of an impending recession. Conduct a brief research on this topic and explain the phenomenon. And comment on the recent steepening of the yield curve. Share your opinion on the matter. (10 marks)

Data source for yield curve plotting:

Daily Treasury Rates | U.S. Department of the Treasury

Other useful information

Applied Macroeconomics and Econometrics Center - FEDERAL RESERVE BANK of NEW YORK